

Data Scraping Pipeline: Tanqeeb Job Board

This notebook implements an automated web scraping pipeline to collect job market listings from **Tanqeeb Egypt**. The extracted data covers multiple tech and data-related domains to analyze current hiring trends, skill demands, and job market dynamics.

Technical Features & Execution Logic

- **Target Categories & Search Scope:** Dynamic pagination across seven key data domains:
 - Data Analysis (296 pages)
 - Data Science (131 pages)
 - Data Engineering (231 pages)
 - Business Intelligence (108 pages)
 - Machine Learning (43 pages)
 - Data Entry (38 pages)
 - Artificial Intelligence (30 pages)
- **Session Persistence (`requests.Session`):** Reuses TCP connections across loop iterations to optimize performance and lower request overhead.
- **Custom User-Agent Spoofing:** Standardized browser headers applied to mimic legitimate user traffic and prevent connection blocking.
- **Rate Limiting & Server Respect:** Includes explicit sleep intervals (`0.5s`) between page loads to ensure non-disruptive scraping.
- **Error Handling & Resilience:** Encapsulated in try-except blocks with retry delays (`3s`) to gracefully manage network timeouts or response issues.
- **Structured Data Extraction:** Captures essential metadata including Title, Company Name, Location, Work Type, Publication Date, Source, and Direct Job Links.

```
In [3]: import requests
from bs4 import BeautifulSoup
import time
import pandas as pd

Jobs_tanqeeb = []

categories = {"Data Analysis": {"query": "Data+Analysis", "pages": 296}, "Data Scie

headers = {"User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537

session = requests.Session()
session.headers.update(headers)

for category, info in categories.items():
```

```

query = info["query"]
pages = info["pages"]

print(f"\n===== {category} | {pages} Pages =====\n")

for page_number in range(1, pages + 1):

    url = f"https://egypt.tanqeeb.com/jobs/search?keywords={query}&page_no={page_number}"

    try:

        response = session.get(url, timeout=30)
        response.raise_for_status()

        soup = BeautifulSoup(response.text, "html.parser")

        jobs = soup.find_all("div", class_="search-job-card")

        print(f"{category} | Page: {page_number}/{pages} | Jobs: {len(jobs)}")

        for i in jobs:

            title = i.select_one("a.search-job-title-link")
            title = title.get_text(strip=True) if title else None

            type_location = i.select_one("span.search-job-workplace-location")
            type_location = type_location.get_text(strip=True) if type_location else None

            date = i.select_one("span.search-job-date")
            date = date.get_text(strip=True) if date else None

            site_category = i.select_one("div.search-job-categories")
            site_category = site_category.get_text(strip=True) if site_category else None

            company_name = i.select_one("div.search-job-company-name")
            company_name = company_name.get_text(strip=True) if company_name else None

            type_work = i.select_one("span.search-job-tag")
            type_work = type_work.get_text(strip=True) if type_work else None

            source = i.select_one("span.search-job-source")
            source = source.get_text(strip=True) if source else None

            location_element = i.select_one("div.search-job-company-city span")
            location = location_element.get_text(strip=True) if location_element else None

            job_link = i.select_one("a.search-job-title-link")
            job_link = "https://egypt.tanqeeb.com" + job_link.get("href") if job_link else None

            Jobs_tanqeeb.append({"Title": title, "Type_Location": type_location, "Date": date, "Site_Category": site_category, "Company_Name": company_name, "Type_Work": type_work, "Source": source, "Location": location, "Job_Link": job_link})

            time.sleep(0.5)

    except requests.exceptions.RequestException as e:

```

```
        print(f"ERROR | {category} | Page: {page_number} | {e}")

        time.sleep(3)

        continue

df = pd.DataFrame(Jobs_tanqeeb)

print("\n=====")
print("Scraping Finished")
print("Total Jobs:", len(df))
print("=====")

print("\nJobs per Data Category:")
print(df["Data_Category"].value_counts())

df.head()
```

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Data Engineering	Page: 209/231	Jobs: 20
Data Engineering	Page: 210/231	Jobs: 20
Data Engineering	Page: 211/231	Jobs: 20
Data Engineering	Page: 212/231	Jobs: 20
Data Engineering	Page: 213/231	Jobs: 20
Data Engineering	Page: 214/231	Jobs: 20
Data Engineering	Page: 215/231	Jobs: 20
Data Engineering	Page: 216/231	Jobs: 20
Data Engineering	Page: 217/231	Jobs: 20
Data Engineering	Page: 218/231	Jobs: 20
Data Engineering	Page: 219/231	Jobs: 20
Data Engineering	Page: 220/231	Jobs: 20
Data Engineering	Page: 221/231	Jobs: 20
Data Engineering	Page: 222/231	Jobs: 20
Data Engineering	Page: 223/231	Jobs: 20
Data Engineering	Page: 224/231	Jobs: 20
Data Engineering	Page: 225/231	Jobs: 20
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Data Engineering	Page: 227/231	Jobs: 20
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Data Entry	Page: 6/38	Jobs: 20
Data Entry	Page: 7/38	Jobs: 20
Data Entry	Page: 8/38	Jobs: 20
Data Entry	Page: 9/38	Jobs: 20
Data Entry	Page: 10/38	Jobs: 20
Data Entry	Page: 11/38	Jobs: 20
Data Entry	Page: 12/38	Jobs: 20
Data Entry	Page: 13/38	Jobs: 20
Data Entry	Page: 14/38	Jobs: 20
Data Entry	Page: 15/38	Jobs: 20

Data Entry	Page: 16/38	Jobs: 20
Data Entry	Page: 17/38	Jobs: 20
Data Entry	Page: 18/38	Jobs: 20
Data Entry	Page: 19/38	Jobs: 20
Data Entry	Page: 20/38	Jobs: 20
Data Entry	Page: 21/38	Jobs: 20
Data Entry	Page: 22/38	Jobs: 20
Data Entry	Page: 23/38	Jobs: 20
Data Entry	Page: 24/38	Jobs: 20
Data Entry	Page: 25/38	Jobs: 20
Data Entry	Page: 26/38	Jobs: 20
Data Entry	Page: 27/38	Jobs: 20
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Data Entry	Page: 29/38	Jobs: 20
Data Entry	Page: 30/38	Jobs: 20
Data Entry	Page: 31/38	Jobs: 20
Data Entry	Page: 32/38	Jobs: 20
Data Entry	Page: 33/38	Jobs: 20
Data Entry	Page: 34/38	Jobs: 20
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Business Intelligence	Page: 5/108	Jobs: 20
Business Intelligence	Page: 6/108	Jobs: 20
Business Intelligence	Page: 7/108	Jobs: 20
Business Intelligence	Page: 8/108	Jobs: 20
Business Intelligence	Page: 9/108	Jobs: 20
Business Intelligence	Page: 10/108	Jobs: 20
Business Intelligence	Page: 11/108	Jobs: 20
Business Intelligence	Page: 12/108	Jobs: 20
Business Intelligence	Page: 13/108	Jobs: 20
Business Intelligence	Page: 14/108	Jobs: 20
Business Intelligence	Page: 15/108	Jobs: 20
Business Intelligence	Page: 16/108	Jobs: 20
Business Intelligence	Page: 17/108	Jobs: 20
Business Intelligence	Page: 18/108	Jobs: 20
Business Intelligence	Page: 19/108	Jobs: 20
Business Intelligence	Page: 20/108	Jobs: 20
Business Intelligence	Page: 21/108	Jobs: 20
Business Intelligence	Page: 22/108	Jobs: 20
Business Intelligence	Page: 23/108	Jobs: 20
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Business Intelligence	Page: 25/108	Jobs: 20
Business Intelligence	Page: 26/108	Jobs: 20
Business Intelligence	Page: 27/108	Jobs: 20
Business Intelligence	Page: 28/108	Jobs: 20
Business Intelligence	Page: 29/108	Jobs: 20
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Business Intelligence	Page: 89/108	Jobs: 20
Business Intelligence	Page: 90/108	Jobs: 20
Business Intelligence	Page: 91/108	Jobs: 20
Business Intelligence	Page: 92/108	Jobs: 20
Business Intelligence	Page: 93/108	Jobs: 20
Business Intelligence	Page: 94/108	Jobs: 20
Business Intelligence	Page: 95/108	Jobs: 20
Business Intelligence	Page: 96/108	Jobs: 20
Business Intelligence	Page: 97/108	Jobs: 20
Business Intelligence	Page: 98/108	Jobs: 20
Business Intelligence	Page: 99/108	Jobs: 20
Business Intelligence	Page: 100/108	Jobs: 20
Business Intelligence	Page: 101/108	Jobs: 20
Business Intelligence	Page: 102/108	Jobs: 20
Business Intelligence	Page: 103/108	Jobs: 20
Business Intelligence	Page: 104/108	Jobs: 20
Business Intelligence	Page: 105/108	Jobs: 20
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Machine Learning	Page: 6/43	Jobs: 20
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Machine Learning	Page: 10/43	Jobs: 20
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Machine Learning	Page: 16/43	Jobs: 20
Machine Learning	Page: 17/43	Jobs: 20
Machine Learning	Page: 18/43	Jobs: 20
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Machine Learning	Page: 27/43	Jobs: 20
Machine Learning	Page: 28/43	Jobs: 20
Machine Learning	Page: 29/43	Jobs: 20
Machine Learning	Page: 30/43	Jobs: 20
Machine Learning	Page: 31/43	Jobs: 20

Machine Learning	Page: 32/43	Jobs: 20
Machine Learning	Page: 33/43	Jobs: 20
Machine Learning	Page: 34/43	Jobs: 20
Machine Learning	Page: 35/43	Jobs: 20
Machine Learning	Page: 36/43	Jobs: 20
Machine Learning	Page: 37/43	Jobs: 20
Machine Learning	Page: 38/43	Jobs: 20
Machine Learning	Page: 39/43	Jobs: 20
Machine Learning	Page: 40/43	Jobs: 20
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Artificial Intelligence	Page: 12/30	Jobs: 20
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Artificial Intelligence	Page: 14/30	Jobs: 20
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Artificial Intelligence	Page: 16/30	Jobs: 20
Artificial Intelligence	Page: 17/30	Jobs: 20
Artificial Intelligence	Page: 18/30	Jobs: 20
Artificial Intelligence	Page: 19/30	Jobs: 20
Artificial Intelligence	Page: 20/30	Jobs: 20
Artificial Intelligence	Page: 21/30	Jobs: 20
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Artificial Intelligence	Page: 23/30	Jobs: 20
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Artificial Intelligence	Page: 25/30	Jobs: 20
Artificial Intelligence	Page: 26/30	Jobs: 20
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Artificial Intelligence	Page: 28/30	Jobs: 20
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Artificial Intelligence	Page: 30/30	Jobs: 20

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Scraping Finished

Total Jobs: 16969

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Jobs per Data Category:

Data_Category	
Data Analysis	5379
Data Engineering	4617
Data Science	2620

```
Business Intelligence      2157
Machine Learning          852
Data Entry                 744
Artificial Intelligence    600
Name: count, dtype: int64
```

Out[3]:

	Title	Type_Location	Date	Category	Data_Category	Company_name	Type_W
0	Statistics And Data Analysis Trainer	Hybrid - Egypt - Cairo	13 August 2026	Education & Training	Data Analysis	E3mel business Academy	Full T
1	Data Analysis Specialist	On-site - Egypt - Cairo	26 July 2026	Information Technology	Data Analysis	j&t express	Full T
2	Data Analysis and Revenue Growth Specialist	On-site - Egypt - Cairo	5 August 2026	Accounting, Finance & Banking	Data Analysis	geidea	Full T
3	Senior Data Engineer - AI Business Analysis	On-site - Egypt - Cairo	7 July 2026	Logistics, Supply Chain & Procurement	Data Analysis	NOK Human Capital	Full T
4	Content Marketing & Data Analysis Assistant	On-site - Egypt - Cairo	17 July 2026	Sales	Data Analysis	Wecrin Inc.	Interns

Data Export & Storage

Saves the fully aggregated jobs dataset into a cleaned CSV file for downstream data cleaning, preprocessing, and analytical workflows.

- **File Format:** CSV (`.csv`)
- **Encoding:** `utf-8-sig` (ensures proper display of Arabic characters and special symbols in Excel)
- **Index Exclusion:** `index=False` (prevents exporting pandas automatic row indices)

```
In [16]: df.to_csv(r'D:\Tanqeeb_Jobs.csv', index=False, encoding='utf-8-sig')
```